

Laboratory Problem: Student Performance Analysis Using K-Means Clustering

Problem Statement

The Student Affairs Office of ABC University has collected academic and behavioral data from **1,000 students**. The university administration wants to identify groups of students with similar characteristics to better design academic support programs, scholarship opportunities, and intervention strategies.

Since there are no predefined categories indicating whether a student is high-performing, average-performing, or at-risk, the university has decided to use **K-Means Clustering**, an unsupervised machine learning algorithm, to discover natural groupings within the student population.

You have been hired as a **Junior Data Analyst** to analyze the dataset and determine meaningful student clusters based on the following factors:

- Hours Studied Per Week
- Attendance Percentage
- Assignments Completed
- Quiz Average
- Project Score
- Internet Usage Hours
- Extracurricular Activities

Your task is to preprocess the data, determine the optimal number of clusters using the Elbow Method, apply K-Means clustering, visualize the results, and provide recommendations based on the discovered student groups.

Objectives

Using the provided dataset (STUDENT-KMEANS.csv), perform the following tasks:

1. Load and explore the dataset.
2. Identify the features to be used for clustering.
3. Standardize the data.
4. Determine the optimal number of clusters using the Elbow Method.
5. Apply K-Means Clustering.
6. Assign cluster labels to each student.
7. Visualize the clusters.
8. Analyze the characteristics of each cluster.
9. Provide recommendations for each student group.

Tasks

Part A: Data Exploration

1. Load the dataset into Python.
2. Display the first five records.
3. Identify the number of rows and columns.

4. Check for missing values.
5. Generate summary statistics.

Part B: Determining the Number of Clusters

1. Standardize the selected features.
2. Apply the Elbow Method for values of K from 1 to 10.
3. Generate the Elbow Method graph.
4. Identify the optimal value of K.

Part C: K-Means Clustering

1. Train a K-Means model using the chosen value of K.
2. Assign cluster labels to all students.
3. Add a new column named **Cluster** to the dataset.
4. Save the clustered dataset.

Part D: Cluster Analysis

For each cluster:

1. Compute the average values of all features.
2. Describe the characteristics of students in the cluster.
3. Determine whether the cluster represents:
 - High-performing students
 - Average-performing students
 - Students needing academic intervention

Part E: Recommendations

Provide at least **three recommendations** for the university based on the clustering results.

Examples:

- Academic tutoring programs
- Scholarship opportunities
- Time-management workshops
- Attendance monitoring systems